OIDD 6110 · Operations: Quality & Productivity · one-page sheet (every HW1 question type, formula + worked example)

① 1 RESOURCE: flow rate · utilization · cycle time

Capacity = m / p (×60 for /hr) · **Flow rate R = min(Demand, Capacity)** · **Utilization = R / Capacity** (≤100%) · **Cycle time CT = 1 / R**

Demand < cap → demand-constrained; else capacity-constrained (bottleneck 100%).

> Laser cutter cap 5 jobs/hr, demand 4/hr → R = min(4,5) = **4/hr**; u = 4/5 = **80%**; CT = 1/4 hr = **15 min/job** **(not 60/5 = 12)**

② MANY STEPS / PARALLEL UNITS: bottleneck · empty system

Box = resource (worker/machine, add up its activities), △ = inventory.

$Cap_i = m_i/p_i \times 60$ → **Bottleneck = lowest cap** = process capacity **(not longest p!)**

Time for X units from EMPTY = Σp + (X−1) × CT, CT = 60/process cap (min)

> Circuit: Bike 8 units·5 min = 96/hr; Rope 6·4 = 90; Treadmill 15·10 = 90; Kettlebell 12·9 = **80**; Box 10·6 = 100 → cap **80/hr**. 60 athletes from empty: Σp = 34; CT = 0.75 min → 34 + 59×0.75 = **78.25 min**

"Which action raises capacity most?" Recompute every option; only bottleneck changes help, until the next station limits.

> +1 kettlebell → 13/9×60 = 86.7 → cap 86.7 · kettlebell 9→8 min → 12/8×60 = 90 → cap **90 (best)** · +1 treadmill → still 80 · faster bike → still 80

Parallel (vs sequential) work: ↓ flow time only, same capacity. Work cell (each of 4 does all 6.5 min): cap 4/6.5×60 = 36.9/hr.

③ LABOR: labor content · labor utilization · idle

Labor content LC = Σ p_i (each step once; staffing irrelevant)

Avg labor utilization = LC / (CT × total # workers)

$Idle_i$ per cycle = CT × m_i − p_i · CT = 1/R (if demand-constrained CT = 1/demand!)

> Passport office: Intake 3 staff·12 min (15/hr), Review 4·20 (**12/hr**), Photo 1·4 (15/hr) → cap 12/hr; LC = 12+20+4 = **36 min**; CT = 5 min, 8 workers → 36/(5×8) = **90%**

> Subway 37/46/37 s: CT 46; idle 9+0+9 = 18 s; LC 120; 120/(46×3) = 87%

④ COST · TAKT · MANPOWER · ASSIGN WORKERS

Cost of direct labor = total wages per hr / R

(R = min(demand, cap): includes idle time)

Takt = available time / demand · **Target manpower = LC / takt** (lower bound; round up only if asked)

> Steps 8, 5, 6 min, 1 worker each, $15/hr. Demand 6/hr: cap 60/8 = 7.5 > 6 → R = 6 → $45/6 = **$7.50/unit** **(not $45/7.5)**. Demand 15/hr: takt = 60/15 = **4 min**; LC 19 → TM = 19/4 = **4.75**

N workers to assign (each step ≥1): add one at a time to the current bottleneck, check combos.

> 5 workers on 8/5/6: (2,1,2) → 15, 12, 20 → **12/hr**; (2,2,1) → 10; (3,1,1) → 10

> Mr K: S1 10, S2 20, S3 30 min → bottleneck S3 2/hr; u(S2) = 2/3; LC 60; ALU = 60/(30·3) = 67%; $20×3/2 = $30; 10 guests empty = 60+9×30 = 330 min; takt 600/40 = 15; TM 4

⑤ LITTLE'S LAW · INVENTORY TURNS · COST

I = R × T (same time units) · T = I/R

> ER 180 patients/day, 60 inside → T = 60/180 day = **8 h**

Turns = COGS / Inventory **(not revenue)** · **Days of supply = Inv/COGS × 365** = 365/turns

Per-unit inventory cost = (annual %) / turns × item cost

> Rev 3,450, COGS 2,070, Inv 345 → turns 2,070/345 = **6**; 24%/6 = 4% → $150 item: **$6.00**; days = 365/6 = 60.8

True: higher turns ⇒ shorter time in inventory ⇒ lower per-unit holding cost. **Not implied:** higher revenue, margin or profit (Tiffany 0.66 turns, 61% margin; Kroger 13.4). Compare competitors only.

Turnover = leavers/yr ÷ employees · tenure at departure = 1/turnover · **avg tenure = ½ × tenure at departure**

> Costco 160k staff, avg 9 yrs → 18 yrs at departure; turnover 1/18 = 5.6%; hires = 160,000/18 = 8,889/yr

⑥ BUILD-UP DIAGRAM (inventory / backlog over time)

Slope = supply rate − demand rate (+ = inventory, − = backlog)

Convert:

demand/hr = customers × units each / hours open

Last customer served = start + total units / production rate (if backlog at end)

Level at time t = P·(t − start) − D·(t − open); max backlog = level when arrivals stop

Runs out when P(t − s) = D(t − open) → **t = (D·open − P·s)/(D − P)**

> Juice bar: 120 customers 7am–1pm × 2 bottles = 240 → D = 40/hr; pressing 30/hr from 6am. 7am: +30. Net −10/hr → out at 10am; 1pm: −30 → **max backlog 30**; last served 6 + 240/30 = **2pm**. Check: produced = demanded ✔

> Donut: bake 60/hr from 4am, sell 120/hr 7–11am → 180 at 7am; 0 at 10am; −60 at 11am; served by noon

⑦ MULTIPLE FLOW UNITS (product mix) · ATTRITION

1) Demand at resource = total demand × Σ mix% of types that visit it

2) cap = m/p 3) **Implied util = demand / cap** (can be >100%) → **bottleneck = HIGHEST implied util** **(not lowest capacity)**

Max demand (same mix) = total demand / max implied util

· minutes-of-work: $\Sigma(d_k \cdot p_k)$ / (m·60)

> Taco shop 5 orders/min: 30% burrito, 50% bowl, 20% drink. Assemble (burrito+bowl) m=2, 0.4 min: demand 5×0.8 = 4/min; cap 2/0.4 = 5 → **80%**. Grill (burrito) 1, 0.5: 1.5/2 = 75%. Prep 4/6 = 67%, Drinks 1/2 = 50%, Pay 5/6.67 = 75% → bottleneck **Assemble**; max demand = 5/0.8 = **6.25/min**

> Prof: File 18/20 = 90%, Foreign 3/6 = 50%, Dept1 14/12 = **117%**, Dept2 4/15, Print 18/30 → Dept1; max 18/1.167 = 15.4/hr

Process yield = $y_1 \cdot y_2 \ldots$ · **start = Q / yield** · downstream demand = arrivals × earlier yields

> Capital One 26.4% booked → 700/0.264 = 2,652 apps; UW 130/day vs cap 8×400/40 = 80 → 163% bottleneck

⑧ WAITING (variability) · TRANSIENT QUEUE

a = avg time BETWEEN arrivals (30/hr → a = 2 min) · **u = p/(a·m)** (<1) · **CV = sd/mean** (Poisson CVa = 1)

$Tq = (p/m) \cdot u^{\sqrt{(2(m+1))}-1}/(1-u) \cdot (CVa^2+CVp^2)/2$; m=1: p·u/(1−u)·(…)/2

T = Tq + p · **Iq = Tq/a** · Ip = p/a · I = Iq + Ip

> Emails a=2 (sd 2), p=4 (sd 2), m=3: u = 0.667; CVa 1, CVp 0.5 → 0.625; $(4/3)\cdot 0.667^{1.83}/0.333 = 1.43$ → Tq = **1.19 min**, T = 5.19, Iq = 0.6

u/(1−u): 80%→4, 90%→9, 95%→19. u ≥ 100% → **triangle**: longest queue = (in − out)×duration; **total wait = ½ × (build-up + depletion) × longest queue**; avg = total / # customers

> Brewery 2/min in, 1/min out, 60 min → queue 60; depletes in 60 min; total ½·120·60 = 3,600 cust-min; /120 = **30 min**

⑨ LEAN · LOSS · QUALITY · INTRO (concepts)

OEE = value-add time / paid time (resource) · **%VAT = value-add / flow time** (flow unit). MRI 18×40 min = 12h / 20h = 60%; radiology visit 20/135 = 15%. Push piles WIP before the bottleneck (20→10→15/hr); pull/kanban caps WIP (cards); JIT; heijunka needs short setups; jidoka = detect-stop-alert (Andon); poka-yoke = fool-proof.

Loss: **$r = p/a$; $P_m(r) = (r^m/m!)/\Sigma_{0..m}\, r^k/k!$**; R = $(1-P_m)/a$. r=2, m=3 → 21.1%. Pooling ↓ waits/loss.

Quality: **$Cp = (USL-LSL)/6\sigma$**; six sigma = Cp 2; ±3σ → 0.27% defects (two tails).

Intro: frontier = firms not Pareto-dominated (no one better on all dims); inhibitors = waste, variability, inflexibility (volume vs mix flexibility).

✔ CHECKLIST (1 pt, no partial credit)

Units (/min vs /hr; ×60) · is demand < capacity? · CT = 1/R · labor cost uses R · implied util for mixes · customers→units · keep 3 decimals · never blank.